External Validation of SpineNetv2 Deep Learning System for Automated Lumbar Spine MRI Analysis

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Inventory Category:	External validation / generalizability
Comparability / Priority:	High | Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	External validation | Lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Multiple lumbar pathologies + Pfirrmann grading
Model / AI Method:	SpineNetv2
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Recent independent validation emphasizing that binary pathology detection and ordinal grading may generalize differently.

Abstract / Summary

Abstract Background Magnetic resonance imaging (MRI) is the reference standard for evaluating degenerative lumbar spine disorders, but interpretation is time-consuming and subject to inter-observer variability. SpineNetv2, a publicly available deep learning system, enables automated analysis of multiple spinal pathologies. This study conducted an independent external validation of SpineNetv2 against expert reference standards. Methods A total of 491 patients (2,455 lumbar discs, L1/2–L5/S1) were retrospectively included. Disc-level reference labels were established by an expert orthopedic surgeon, with a junior orthopedic surgeon serving as comparator. Six pathologies were assessed: disc degeneration (Pfirrmann grading), central canal stenosis (CCS), spondylolisthesis, herniation, and bilateral foraminal stenosis (FS). Performance metrics included sensitivity, specificity, positive predictive value, negative predictive value, F1-score, Matthews correlation coefficient, exact agreement, weighted kappa, and mean absolute error. McNemar's test and bootstrap resampling (1,000 iterations) were used for statistical analysis. Results Overall accuracy ranged from 83.5–97.5% (mean 92.8%). SpineNetv2 significantly outperformed the junior orthopedic surgeon in CCS, spondylolisthesis, and bilateral FS (all $p \leq 0.001$), with comparable performance in herniation ($p = 0.293$). Pfirrmann grading showed lower MAE for SpineNetv2 compared with the junior surgeon (0.213 vs. 0.254, $p = 0.001$), though accuracy declined in older patients and upper lumbar discs. Error analysis revealed a specificity-oriented profile, with false negatives exceeding false positives. Conclusions SpineNetv2 demonstrated high accuracy across five binary lumbar pathologies, while Pfirrmann grading remained the main limitation, particularly in elderly upper lumbar discs. Its specificity-oriented profile supports use as a confirmatory second reader, but reliance on negative findings is not recommended. Broader reliability will require multicenter, multi-reader validation and sensitivity-oriented calibration.